Alexandria Quantum Hackathon 2026

QML Challenge

Quantum Phase Recognition

Few & Qurious

Meet the Team!



Few & Qurious

Background & Motivation

Real-world quantum measurements destroy states, making data scarce.

Measurement-efficient classification is vital for discovering ne quantum materials and advancing energy technologies

Problem Statement

- The Problem: Classify quantum phases of many-body ground states using a strict limit of 64 copies per test state.
- Must handle noise, unseen system sizes, and unknown "impostor" phases
- Fixed parameter Architecture: The model's trainable parameter count must remain constant and run unmodified regardless of system size.

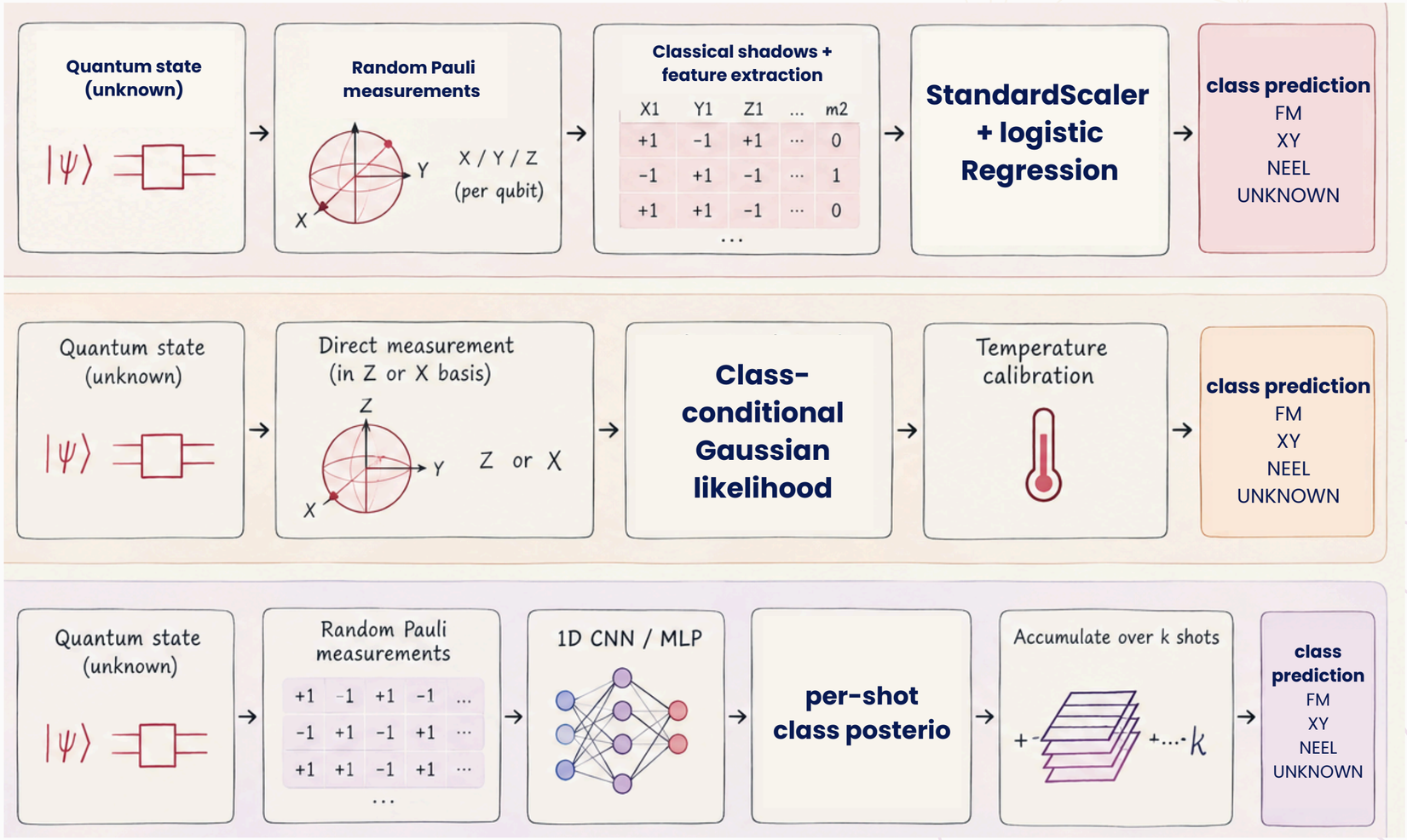
Aim: beat classical baselines under the same budget

Classical Approach

classical shadows+
Logistic Regression

Order-Parameter
Measurement

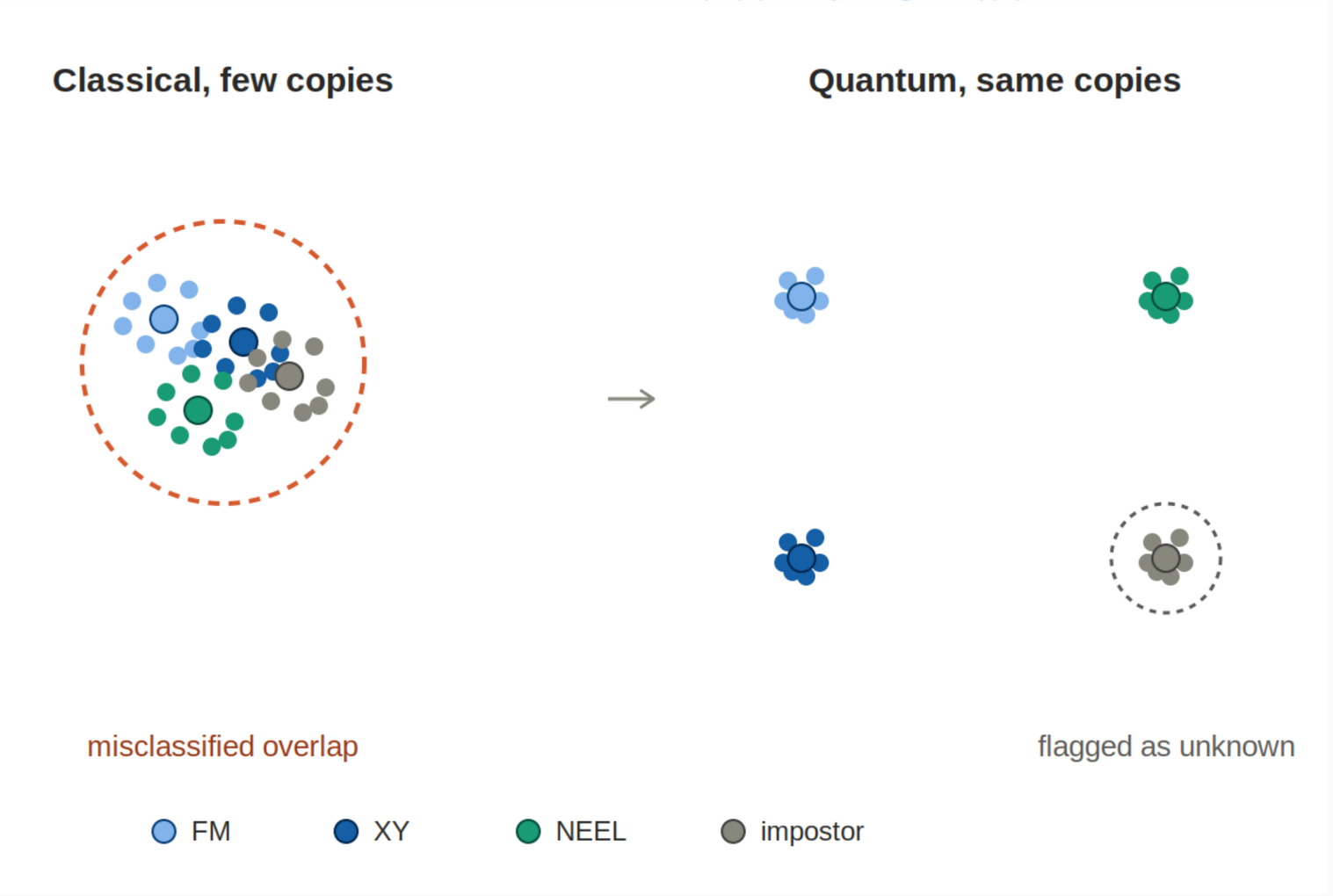
1D CNN on
raw bitstring



Why Quantum ?

- on few copies, classical representations overlap so classes are not separable

- on the same few copies, quantum promises better representations, hence better separation



Proposed Quantum PIPELINE

Train set



Quantum Phase dataset

Correlation-based Adaptive Thresholding

Test Set



Pennylane fermihubbard

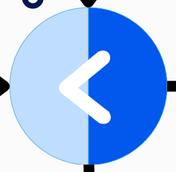


QML Backbone



Measurement

Thresholding



Unknown

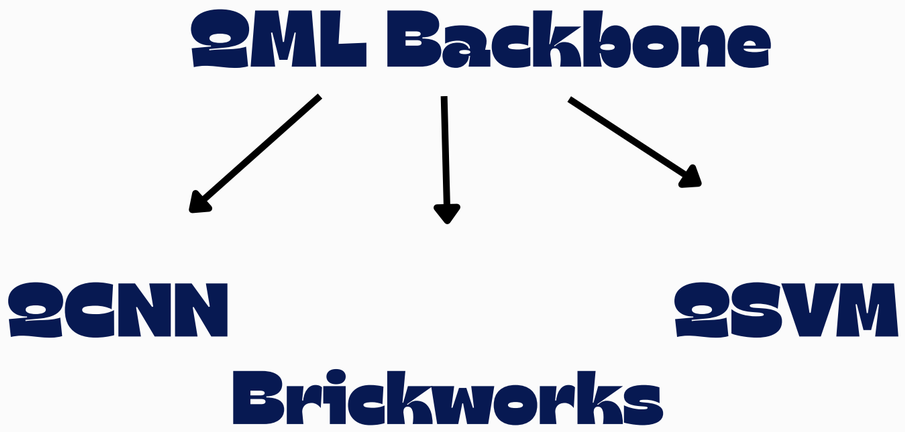
Bayesian Classification Head

FM

XY

NEEL

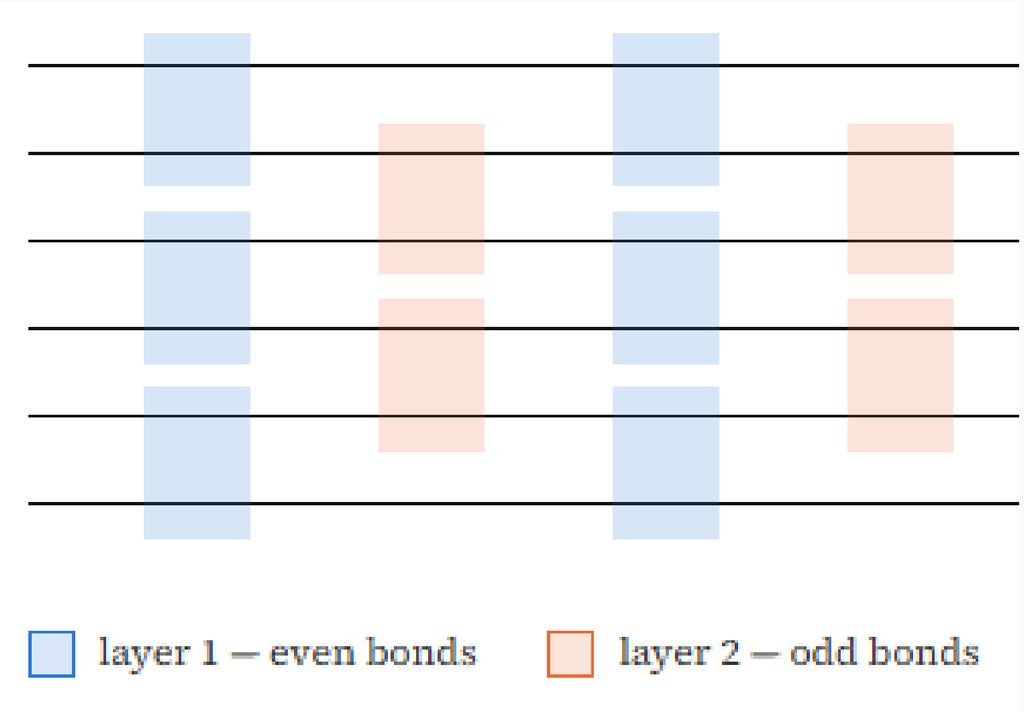
Experiment Design



Noise variations (p)	range [0.01-0.05]
Imposters	Pennylane fermihubbard

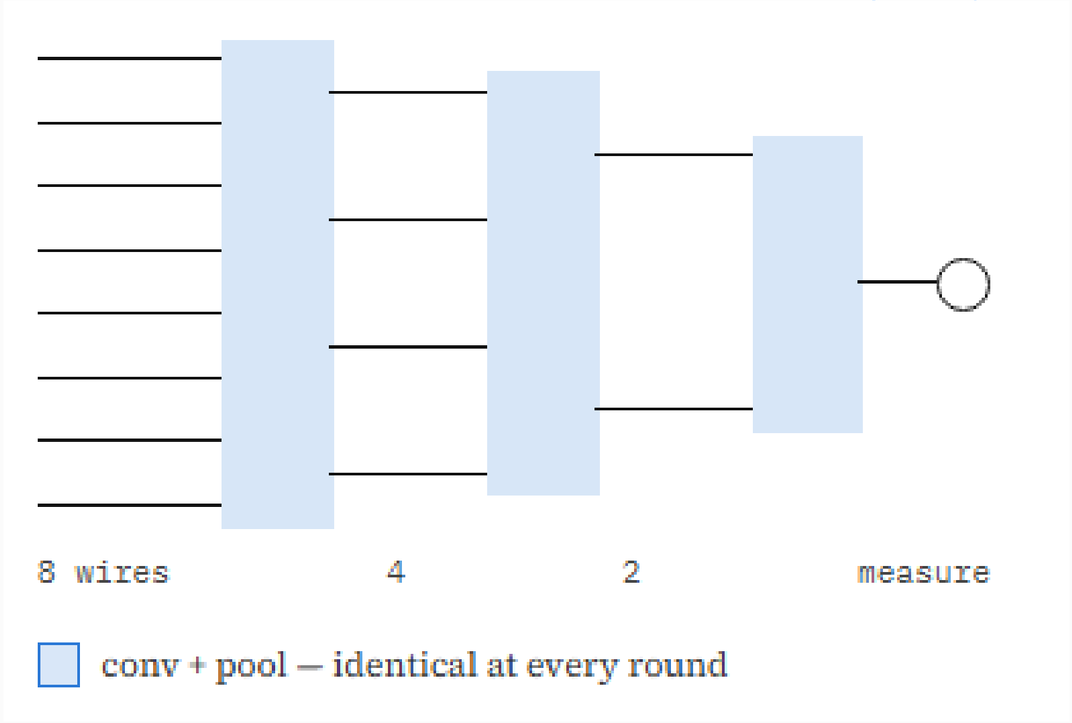
QML Backbone

BRICKWORK



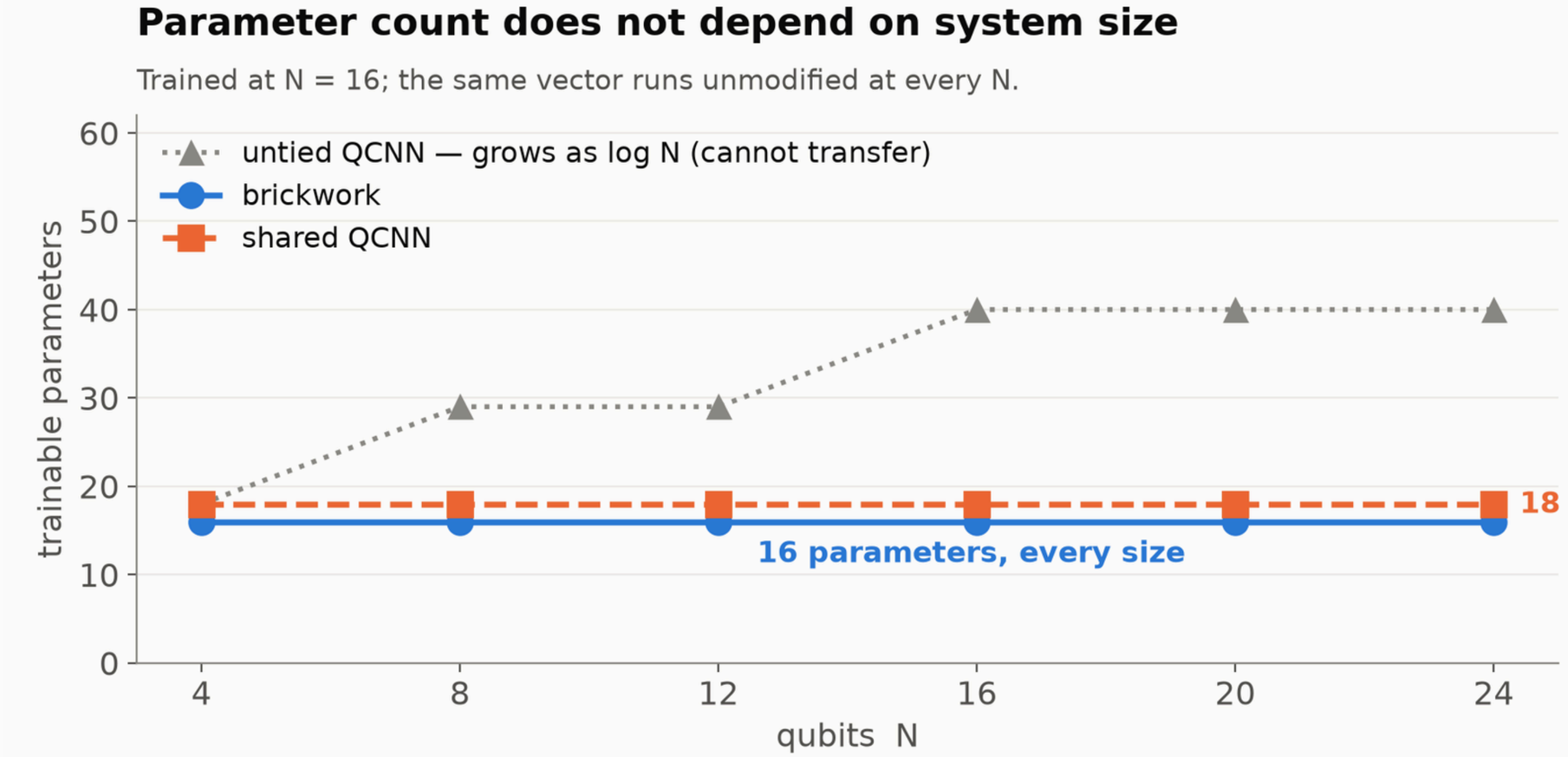
K. Chinzei, Q.H. Tran, K. Maruyama,H. Oshima, and S. Sato,
"Splitting and parallelizing of quantum convolutional neural networks for learning translationally symmetric data," Physical Review Research, vol. 6, art. 023042, 2024, doi: 10.1103/PhysRevResearch.6.023042. [Online]. Available: https://arxiv.org/abs/2306.07331

QCCN

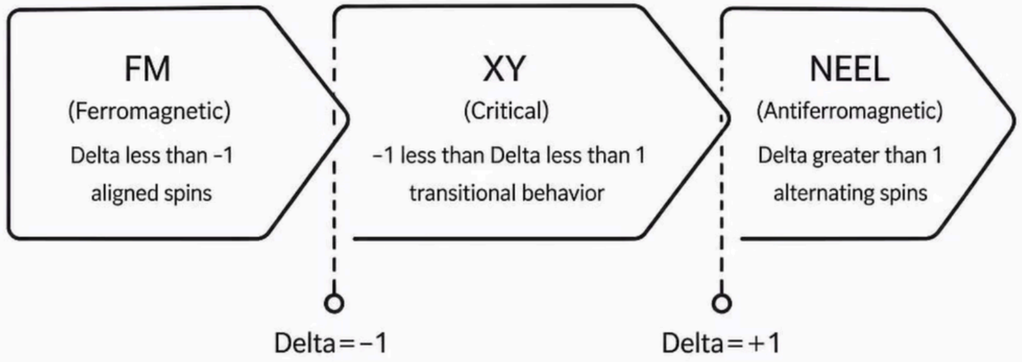


. Cong, S. Choi, and M. D. Lukin, "Quantum convolutional neural networks," Nature Physics, vol. 15, no. 12, pp. 1273-1278, 2019, doi: 10.1038/s41567-019-0648-8. [Online]. Available: https://arxiv.org/abs/1810.03787

QML Backbone



DataSet Overview



Training Data

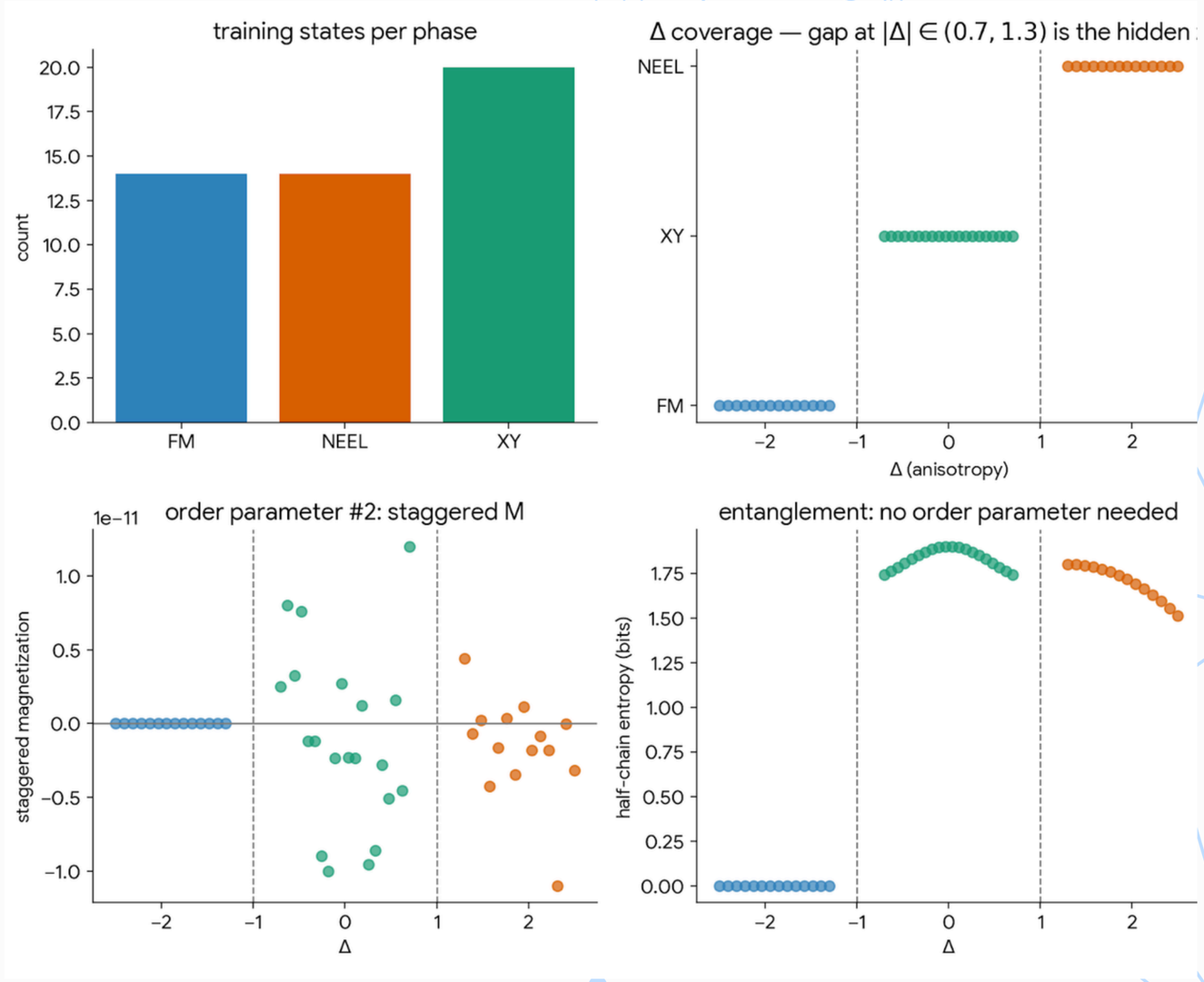
48 labeled states with full measurement access across all three phases

Test Oracle

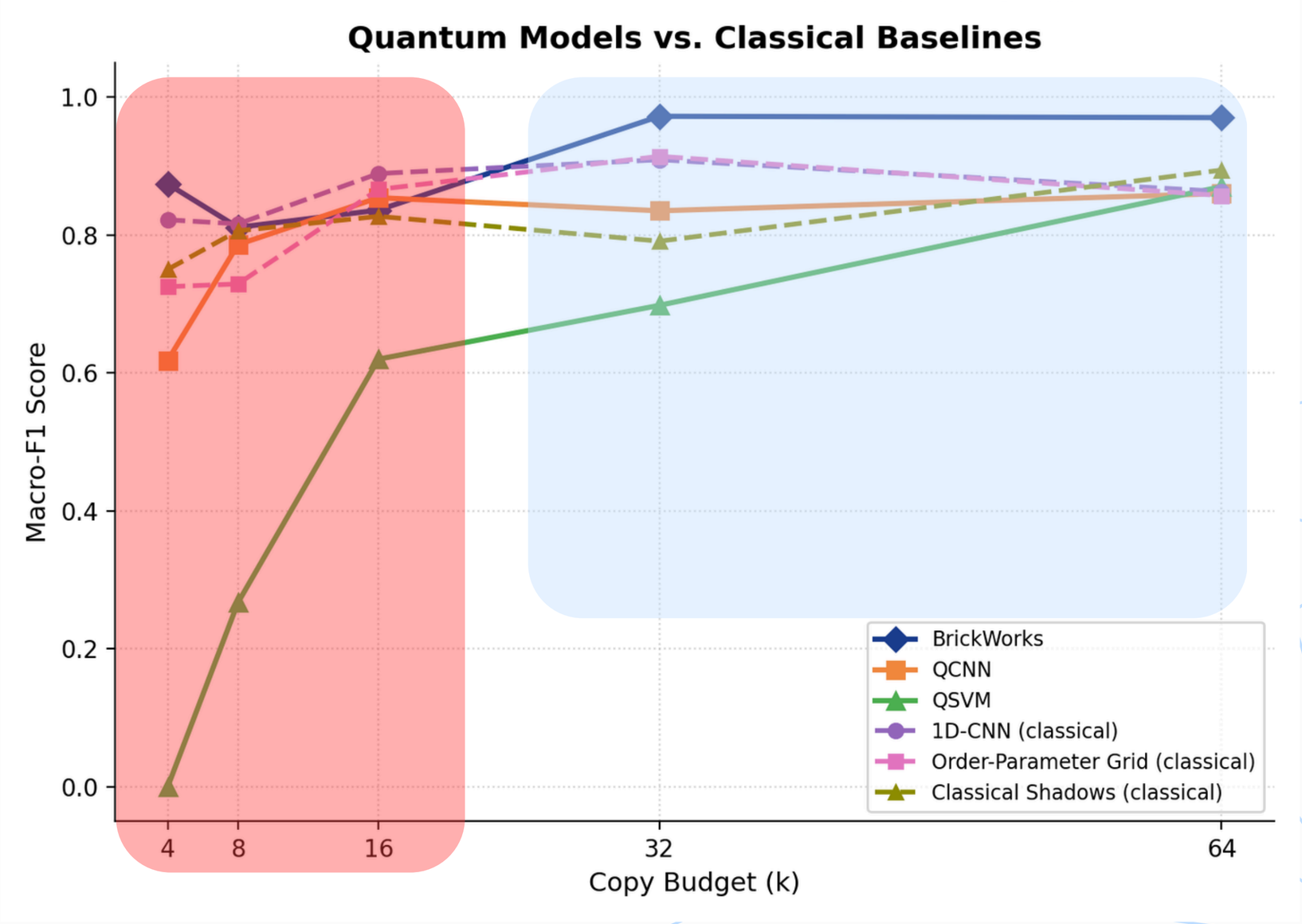
64 copies maximum per unseen state — noisy, metered, unforgiving

Hidden Gap

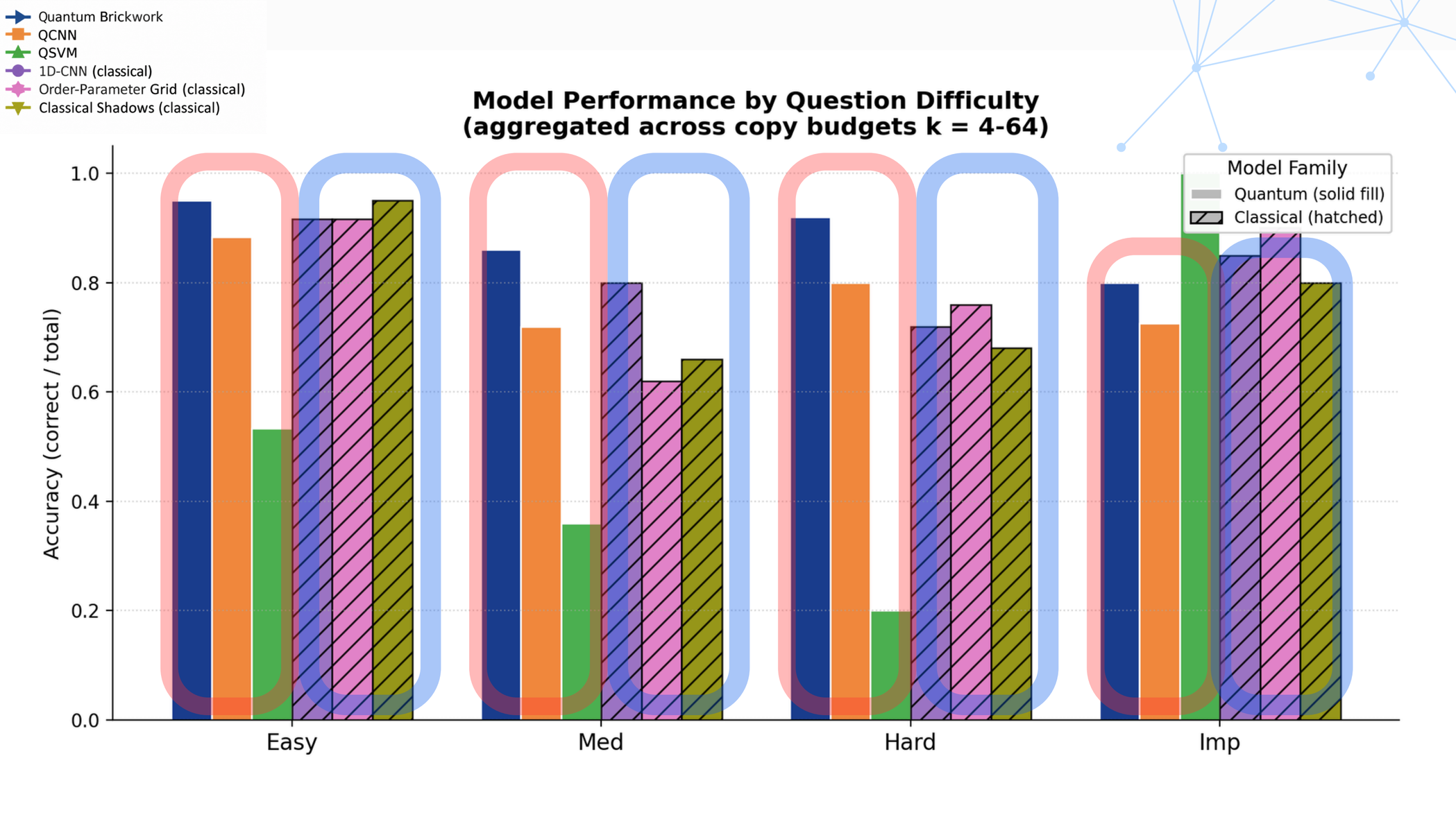
$\Delta \in (0.7, 1.3)$ — the **critical transition zone** is withheld from training



Experimental Results

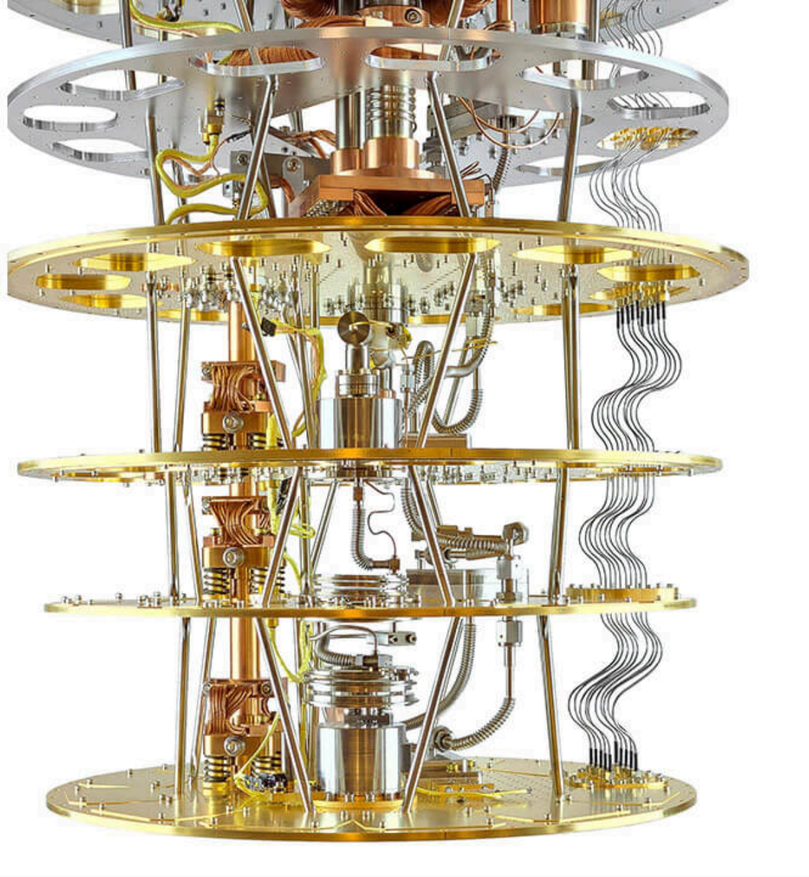


Experimental Results



IBM Quantum Hardware

- ibm_marrakesh backend
- Tested sample from each label
- Used shots = 8



```
--- State H13 (true label: FM) ---
  shot 1/8: bits=[0 1 1 1 0 1 1 0 0 0 1 0 0 0 0 0]  running_prediction(after 1 shots)=XY
  shot 2/8: bits=[0 0 0 0 0 1 1 1 1 0 0 0 0 0 0 1]  running_prediction(after 2 shots)=FM
  shot 3/8: bits=[1 0 1 1 0 1 1 0 0 0 0 0 1 1 1 1]  running_prediction(after 3 shots)=XY
  shot 4/8: bits=[0 0 1 0 1 1 0 0 0 0 0 0 1 1 0 0]  running_prediction(after 4 shots)=XY
  shot 5/8: bits=[0 0 0 0 0 1 1 0 1 1 0 0 1 1 0 0]  running_prediction(after 5 shots)=XY
  shot 6/8: bits=[1 1 0 1 0 0 1 0 0 0 0 1 1 0 0 1]  running_prediction(after 6 shots)=XY
  shot 7/8: bits=[1 0 0 1 0 1 1 1 0 0 0 0 1 0 1 1]  running_prediction(after 7 shots)=XY
  shot 8/8: bits=[0 0 0 0 0 0 0 1 1 0 0 0 1 0 0 0]  running_prediction(after 8 shots)=FM
  FINAL prediction for H13 (k=8): FM   true=FM   CORRECT
```

Conclusion

- on low copies, real separation : quantum beat classical shadows on the same tight budget
- Fixed architecture, zero retraining : generalizes across system sizes out of the box
- Robust to noise, aware of impostors : flags what it doesn't recognize

Thanks

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